

X47.CE Installation Instructions Outside USA

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Introduction

The X47.CE Universal Relay Pack (PN 201-147.CE) is an INNCOM Integrated Room Automation System (IRAS) device designed for guestroom controls. The X47.CE provides five relay outputs used to switch a variety of loads, including fan coil units (FCU) and light-duty general loads. The X47.CE can integrate with and receive data from multiple points throughout the IRAS using low-voltage wired communication (S5bus) or radio frequency (RF) communication (when connected to a PC502). The X47.CE is a 220-240 VAC line power controller.



Figure 1. X47.CE

Features

- Controls a full range of FCU systems, complex light-duty lighting circuits, and other guestroom applications from 24–277VAC
- Supports three communication modes: S5bus, IR, and RF
- Installs inside FCU/HVAC enclosures
- Connects to remote thermistor (04-1096, 04-1096.MOD 04-1095)

for external temperature sensing (+/- 1%temperature accuracy at 25°C)

- Enclosed to protect sensitive circuitry
- Complies with agency-approved recommendations for worldwide usage

X47.CE FCU Controller: In combination with the INNCOM e4 Smart Digital Thermostat, the X47.CE FCU controller is a feature-rich, cost-saving energy management tool. Communication using RF or INNCOM's IR Eye5 Infrared Transceiver adds to its installation versatility. (See the References section of this document for information on INNCOM's Energy Management System).

Safety: The X47.CE is intended for indoor use only. To comply with industry standards in the U.S., Canada, Europe, and Asia, the device is equipped with an enclosure that protects the relays and circuitry from the environmental. The X47.CE can be installed in a NEMA-type electrical box in the field and is permanently wired according to the guidelines of the National Electrical Code. The X47.CE has approval from CE Mark.

Application

The X47.CE has five power relays and a local logic controller card combined in one assembly. Other features include

Blade connectors for 220VAC–240VAC

Four screw openings in the bottom enclosure for mounting in a NEMA enclosure

LED: The X47.CE has one diagnostic LED, which will blink on and off to indicate activity. It also signals during startup and the commissioning process (see below).

Headers: The X47.CE contains the following headers:

H1: In Circuit Programming Header

H2: Digital Input/Output 0V or 5V

H3/4: External temperature sensor headers; Pin 1, 2-GND, Pin 3-10k NTC

H5 IR Eye5 header

H6: 100VAC power and relay output header

H8: Analog output header; Pin 1-GND, Pin 2 Analog output 0–10VDC, Pin 3 Analog output 0–10VDC

H9: 5 pin S5bus I/O header

H10: ES1 header

H11: RS485 adapter header (PN 203-250)

Relays: Tables 1 and 2 show the Relay Output specifications for the X47.CE; Table 3 shows the relay functions for typical applications.

**Table 1: X47.CE Relay Output Specifications,
220-240VAC Input model**

	Rating	Pin Assignment
K1	3A, 277VAC Gen. Purpose 3.6FLA, 21.6LRA	P8
K2	3A, 277VAC, Gen. Purpose, 2.2FLA, 13.2LRA, 0.7A sealed, 7A inrush	P9
K3	3A, 277VAC, Gen. Purpose, 2.2FLA, 13.2LRA, 0.7A sealed, 7A inrush	P10
K4	3A, 277VAC, Gen. Purpose, 2.2FLA, 13.2LRA, 0.7A sealed, 7A inrush	P3
K5	3A, 277VAC, Gen. Purpose, 2.2FLA, 13.2LRA, 0.7A sealed, 7A inrush	P4
Common 1	K1, K2, K3	P5
Common 2	K4, K5	P4

**Table 2. X47.CE Relay Functions for
Typical Installations**

	4 Pipe, 3 Fan, Heat/Cool FCU	2 Pipe, 3 Fan, Cool Only FCU	Heat Pump, 2 Fan 2nd Stage Heat	Heat Pump, 3 Fan Speeds	PTAC, 2 Fan with Heat Strip
K1	High Fan	High Fan	High Fan	High Fan	High Fan
K2	Medium Fan	Medium Fan	W2 2 nd Stage Heat	Medium Fan	Not Connected
K3	Low Fan	Low Fan	Low Fan	Low Fan	Low Fan
K4	Cool Signal	Cool Signal	Y-Compressor	Y-Compressor	Y-Compressor
K5	Heat Signal	Not Connected	B/O-Reversing Valve	B/O-Reversing Valve	W-Heat

Installation

CAUTION!

- The X47.CE installer must be a trained, experienced service technician.
- All wiring must comply with local codes and ordinances.
- The X47.CE is intended for indoor use only.
- Failure to follow the instructions provided with the product could damage the product or cause a hazardous condition.
- The ratings given in the instructions and on the product should be checked to make sure the product is suitable for the application. Table 1 gives relay functions for the line-power version; Table 2 provides relay functions for typical installations.

To install the X47.CE:

Use four 6/32 1/2" sheet metal screws to mount the X47.CE inside a NEMA enclosure or FCU using the four screw holes in the bottom enclosure.

See Figure 2 for line-voltage and 24VAC power supply connectors

See Figure 3 for X47.CE headers

Note: The instructions above describe the typical installation of the X47.CE.

Commissioning

To participate in the network, the X47.CE must be commissioned by teaching it identification information. The following commissioning instructions use an X47.CE connected to a PC502 on the S5bus (for 2.4 GHz wireless RF communications) and an E529 thermostat:

1. Power up the X47.CE (LED blinks rapidly).
2. Enter the Service Mode on the e529 thermostat (press and hold the °F/°C button; press and release the OFF/AUTO button; press and release the DISPLAY button; release the °F/°C button).
3. On the E529 LCD, scroll to **Adr** and press OFF/AUTO.
4. Set the address to teach the X47.CE (typically 14).
5. Press OFF/AUTO. The display will show **bnd**.
6. Use a small point (e.g., the end of a straightened paper clip) to press the Bind switch on the X47.CE once. This sends a Bind request toward the thermostat.
7. The thermostat will send a Bind offer to the X47.CE. If it is accepted, the thermostat will buzz loudly. The X47.CE will reset.
8. Test the X47.CE functionality.

After binding, use the E529 to set Room ID, PAN ID and RF Channel. See property-specific instructions for further guidance.

Wiring

Figure 1 shows the X47.CE line-voltage connectors; Figure 2 shows the headers; and Figure 3 shows a typical connection to a Fan Coil Unit

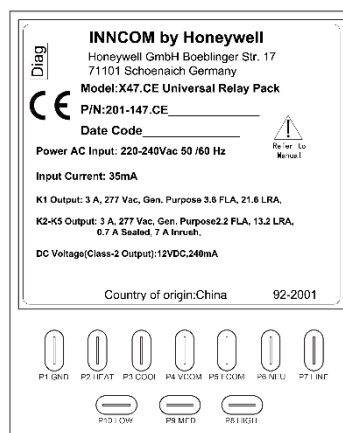


Figure 2 Connectors

Line-Voltage Connections: The phase, neutral, earth, and relay output connections are defined on the product label.

Low-Voltage Power/Communications Connection: The X47.CE can provide 12VDC 240mA output class-2 power and S5bus for IRAS devices.

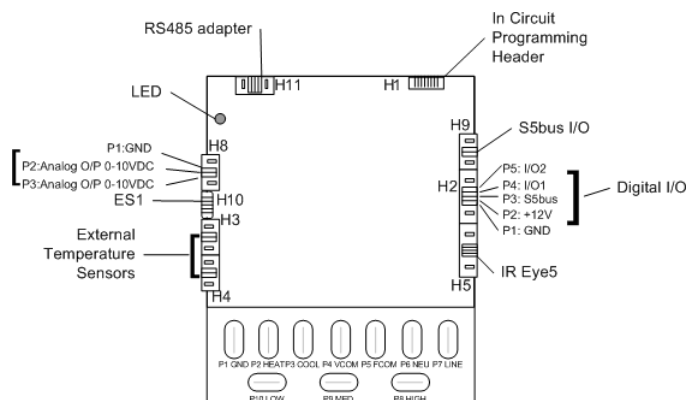


Figure 3 X47.CE headers

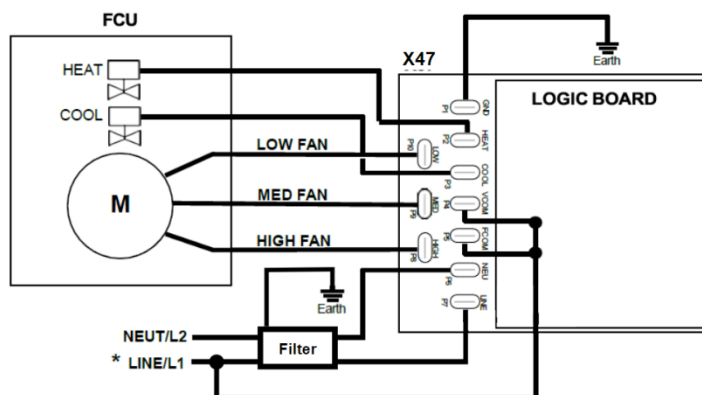


Figure 4 Typical FCU Connections

Specifications

Table 4. Technical Specifications

201-147.CE Class-2 Output Voltage/Power	12VDC, 240mA (See table below to determine final output power)
Parameter Memory	Non-volatile
Temperature range, Operating Storage	32 to 140 degrees F (0 to 40 degrees C) -40 to 158 degrees F (-40 to 70 degrees C) 15 to 99 percent RH
Mounting	4 hole mounting 6/32" bit, 1/2" screw
Dimensions	2 3/4" W x 3 1/2" H (excluding screw tabs) x 1 1/2" D (at deepest part) 70 mm W x 88 mm H x 38 mm D
Approvals	CE Mark, UL, FCC, IC

Current consumption is determined using the following formula: $30\text{mA} + (n \times 20\text{mA})$ where 30mA is the current consumption of the logic board and n= the number of actuated relays.

Table 5. Current Consumption

Configuration	Maximum Load (including logic board)
All relays actuated	130M

Note: Current consumption values of the X47.CE with actuated relays should be deducted from class-2 output power rating in Table 1 in order not to overdrive the power supply. The X47.CE should not be used to provide more than 70mA of external class-2 power.

Model & Part Numbers

Table 6: X47.CE Ordering

Model #	Description
201-147.CE	X47.CE 220-240VAC 50/60Hz w/ 5 relay actuator and Power Supply, HVAC Logic, Grey Housing

Table 7: X47.CE Harness Ordering

Model #	
62-1464.R	X47.CE 24VAC Harness
62-1465.B	X47.CE Low-voltage cable (H3, H4, H8, H9)
62-1511.A	X47.CE Low-voltage cable (H2, H5)

No harness is supplied for the X47.CE with line voltage power supply option.

Table 8: Thermistor Ordering

Model #	
04-1094.MOD	External Thermistor
04-1096	Ruggedized Thermistor

Table 9: RS-485 Adapter Ordering

Model #	
203-250	RS-485 Communication Module

Table 10: Input Filter Ordering

Model #	201-147-FILTER
Description	X47 CONTROLLER INPUT FILTER

FCC Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

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